

Mat 1375 HW5 5.1~5.5

Exercise 5.1

Find $f + g$, $f - g$, $f \cdot g$ for the functions below. State their domain.

✓ a) $f(x) = x^2 + 6x$ and $g(x) = 3x - 5$

✓ b) $f(x) = x^3 + 5$ and $g(x) = 5x^2 + 7$

✓ c) $f(x) = 3x + 7\sqrt{x}$ and $g(x) = 2x^2 + 5\sqrt{x}$

Sol

a) $D_f = (-\infty, \infty)$, $D_g = (-\infty, \infty)$

$$\begin{aligned}(f+g)(x) &= f(x) + g(x) = (x^2 + 6x) + (3x - 5) \\ &= x^2 + 6x + 3x - 5 \\ &= x^2 + 9x - 5\end{aligned}$$

$$\begin{aligned}(f-g)(x) &= f(x) - g(x) = (x^2 + 6x) - (3x - 5) \\ &= x^2 + 6x - 3x + 5 \\ &= x^2 + 3x + 5\end{aligned}$$

$$\begin{aligned}(f \cdot g)(x) &= f(x) \cdot g(x) = (x^2 + 6x) \cdot (3x - 5) \\ &= 3x^3 + 18x^2 - 5x^2 - 30x \\ &= 3x^3 + 13x^2 - 30x\end{aligned}$$

domain

$$\begin{aligned}D_{f+g} &= D_f \cap D_g \\ &= \{x \mid x \in (-\infty, \infty)\}\end{aligned}$$

$$\begin{aligned}D_{f-g} &= D_f \cap D_g \\ &= \{x \mid x \in (-\infty, \infty)\}\end{aligned}$$

$$\begin{aligned}D_{f \cdot g} &= D_f \cap D_g \\ &= \{x \mid x \in (-\infty, \infty)\}\end{aligned}$$

b) $D_f = (-\infty, \infty)$, $D_g = (-\infty, \infty)$

$$\begin{aligned}(f+g)(x) &= f(x) + g(x) = (x^3 + 5) + (5x^2 + 7) \\ &= x^3 + 5x^2 + 12\end{aligned}$$

$$\begin{aligned}(f-g)(x) &= f(x) - g(x) = (x^3 + 5) - (5x^2 + 7) \\ &= x^3 - 5x^2 - 2\end{aligned}$$

domain

$$\begin{aligned}D_{f+g} &= D_f \cap D_g \\ &= \{x \mid x \in (-\infty, \infty)\}\end{aligned}$$

$$\begin{aligned}D_{f-g} &= D_f \cap D_g \\ &= \{x \mid x \in (-\infty, \infty)\}\end{aligned}$$

$$(f \cdot g)(x) = f(x) \cdot g(x) = (x^3 + 5) \cdot (5x^2 + 7) \\ = 5x^5 + 7x^3 + 25x^2 + 35$$

$$D_{f \cdot g} = D_f \cap D_g \\ = \{x \mid x \in (-\infty, \infty)\}$$

c) $D_f = [0, \infty)$, $D_g = [0, \infty)$

(Since $f(x) = 3x + 7\sqrt{x}$ with a square root, then any $x < 0$ will not be in the domain of f , so $D_f = [0, \infty)$.)

Similarly, $g(x)$ has a term of $5\sqrt{x}$, so $D_g = [0, \infty)$.)

$$(f+g)(x) = f(x) + g(x) = (3x + 7\sqrt{x}) + (2x^2 + 5\sqrt{x}) \quad \text{domain} \\ = 2x^2 + 3x + 12\sqrt{x}$$

$$D_{f+g} = D_f \cap D_g \\ = \{x \mid x \in [0, \infty)\}$$

$$(f-g)(x) = f(x) - g(x) = (3x + 7\sqrt{x}) - (2x^2 + 5\sqrt{x}) \\ = -2x^2 + 3x + 2\sqrt{x}$$

$$D_{f-g} = D_f \cap D_g \\ = \{x \mid x \in [0, \infty)\}$$

$$(f \cdot g)(x) = f(x) \cdot g(x) = (3x + 7\sqrt{x}) \cdot (2x^2 + 5\sqrt{x}) \\ = 6x^3 + 14x^2\sqrt{x} + 15x\sqrt{x} + 35(\sqrt{x})^2 \\ = 6x^3 + 14x^2\sqrt{x} + 15x\sqrt{x} + 35x$$

$$D_{f \cdot g} = D_f \cap D_g \\ = \{x \mid x \in [0, \infty)\}$$

Exercise 5.2

Find $\frac{f}{g}$, and $\frac{g}{f}$ for the functions below. State their domain.

✓ a) $f(x) = 3x + 6$

and $g(x) = 2x - 8$

✓ b) $f(x) = x + 2$

and $g(x) = x^2 - 5x + 4$

Sol

a) $D_f = (-\infty, \infty)$, $D_g = (-\infty, \infty)$

Domain

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{3x+6}{2x-8}$$

$$D_{\frac{f}{g}} := D_f \cap D_g \text{ but } g(x) \neq 0$$

$$\Rightarrow (-\infty, \infty) \text{ but } g(x) \neq 0$$

$$\Rightarrow (-\infty, \infty) \text{ but } 2x-8 \neq 0$$

$$\Rightarrow (-\infty, \infty) \text{ but } x \neq 4$$

$$\begin{array}{c} \leftarrow \quad \circ \quad \rightarrow \\ -\infty \quad 4 \quad \infty \end{array}, D_{\frac{f}{g}} = \{x \mid x \in (-\infty, 4) \cup (4, \infty)\}$$

$$\left(\frac{g}{f}\right)(x) = \frac{g(x)}{f(x)} = \frac{2x-8}{3x+6}$$

$$D_{\frac{g}{f}} = D_f \cap D_g \text{ but } f(x) \neq 0$$

$$\Rightarrow (-\infty, \infty) \text{ but } 3x+6 \neq 0$$

$$\Rightarrow (-\infty, \infty) \text{ but } x \neq -2$$

$$\begin{array}{c} \leftarrow \quad \circ \quad \rightarrow \\ -\infty \quad -2 \quad \infty \end{array}, D_{\frac{g}{f}} = \{x \mid x \in (-\infty, -2) \cup (-2, \infty)\}$$

b) $D_f = (-\infty, \infty)$, $D_g = (-\infty, \infty)$

Domain

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x+2}{x^2-5x+4}$$

$$D_{\frac{f}{g}} := D_f \cap D_g \text{ but } g(x) \neq 0$$

$$\Rightarrow (-\infty, \infty), \text{ but } x^2-5x+4 \neq 0$$

$$\Rightarrow (-\infty, \infty), \text{ but } (x-1)(x-4) \neq 0$$

$$\Rightarrow (-\infty, \infty), \text{ but } x-1 \neq 0 \text{ and } x-4 \neq 0$$

$$\Rightarrow (-\infty, \infty), \text{ but } x \neq 1 \text{ and } x \neq 4$$

$$\begin{array}{c} \leftarrow \quad \circ \quad \circ \quad \rightarrow \\ -\infty \quad 1 \quad 4 \quad \infty \end{array}, D_{\frac{f}{g}} = \{x \mid x \in (-\infty, 1) \cup (1, 4) \cup (4, \infty)\}$$

$$\left(\frac{g}{f}\right)(x) = \frac{g(x)}{f(x)} = \frac{x^2 - 5x + 4}{x + 2}$$

$$D_{\frac{g}{f}} := D_g \cap D_f \text{ but } f(x) \neq 0$$

$$\Rightarrow (-\infty, \infty) \text{ but } x + 2 \neq 0$$

$$\Rightarrow (-\infty, \infty) \text{ but } x \neq -2$$

$$\leftarrow \infty \quad \begin{array}{c} \circ \\ -2 \end{array} \rightarrow \infty, D_{\frac{g}{f}} = \{x \mid x \in (-\infty, -2) \cup (-2, \infty)\}$$

Exercise 5.3

Let $f(x) = 2x - 3$ and $g(x) = 3x^2 + 4x$. Find the following compositions:

✓ a) $f(g(2))$

✓ b) $g(f(2))$

✓ c) $f(f(5))$

✓ d) $f(5g(-3))$

e) $g(f(2) - 2)$

f) $f(f(3) + g(3))$

Sol.

a) $g(2) = 3 \cdot (2)^2 + 4 \cdot (2) = 3 \cdot 4 + 8 = 20$

$f(g(2)) = f(20) = 2 \cdot (20) - 3 = 40 - 3 = 37$

b) $f(2) = 2 \cdot (2) - 3 = 4 - 3 = 1$

$g(f(2)) = g(1) = 3 \cdot (1)^2 + 4 \cdot (1) = 7$

c) $f(5) = 2(5) - 3 = 10 - 3 = 7$

$f(f(5)) = f(7) = 2 \cdot (7) - 3 = 14 - 3 = 11$

d) $g(-3) = 3(-3)^2 + 4 \cdot (-3) = 3 \cdot 9 - 12 = 15$

$5g(-3) = 5 \cdot 15 = 75$

$f(5g(-3)) = f(75) = 2(75) - 3 = 150 - 3 = 147$

Exercise 5.4

Find the composition $(f \circ g)(x)$ for the following functions:

✓ a) $f(x) = 3x - 5$ and $g(x) = 2x + 3$

✓ b) $f(x) = x^2 + 2$ and $g(x) = x + 3$

✓ c) $f(x) = x^2 - 3x + 2$ and $g(x) = 2x + 1$

Sol:

a) $(f \circ g)(x) = f(g(x)) = f(2x+3)$ ↙ replace "x" from f(x) by "g(x)"

$$= 3(2x+3) - 5$$

$$= 6x + 9 - 5 = 6x + 4$$

b) $(f \circ g)(x) = f(g(x)) = f(x+3)$

$$= (x+3)^2 + 2$$

$$= x^2 + 6x + 9 + 2 = x^2 + 6x + 11$$

* $(x+3)(x+3)$
 $= x^2 + 3x + 3x + 9$
 $= x^2 + 6x + 9$

c) $(f \circ g)(x) = f(g(x)) = f(2x+1)$

$$= (2x+1)^2 - 3(2x+1) + 2$$

$$= 4x^2 + 4x + 1 - 6x - 3 + 2$$

$$= 4x^2 - 2x$$

* $(2x+1)^2$
 $= (2x+1)(2x+1)$
 $= 4x^2 + 2x + 2x + 1$
 $= 4x^2 + 4x + 1$

Exercise 5.5

Find the compositions

$$(f \circ g)(x), \quad (g \circ f)(x), \quad (f \circ f)(x), \quad (g \circ g)(x)$$

for the following functions:

✓ a) $f(x) = 2x + 4$

and $g(x) = x - 5$

✓ b) $f(x) = x + 3$

and $g(x) = x^2 - 2x$

Sol

a) $(f \circ g)(x) = f(g(x)) = f(x-5)$ → Replace "x" in f(x) by "g(x)"
 $= 2(x-5) + 4 = 2x - 10 + 4 = \boxed{2x - 6}$

$(g \circ f)(x) = g(f(x)) = g(2x+4)$ → Replace "x" in g(x) by "f(x)"
 $= (2x+4) - 5 = \boxed{2x - 1}$

$(f \circ f)(x) = f(f(x)) = f(2x+4)$
 $= 2(2x+4) + 4 = 4x + 8 + 4 = \boxed{4x + 12}$

$(g \circ g)(x) = g(g(x)) = g(x-5)$
 $= (x-5) - 5 = \boxed{x - 10}$

b) $(f \circ g)(x) = f(g(x)) = f(x^2 - 2x)$
 $= (x^2 - 2x) + 3 = \boxed{x^2 - 2x + 3}$

$(g \circ f)(x) = g(f(x)) = g(x+3)$
 $= (x+3)^2 - 2(x+3)$

$$= x^2 + 6x + 9 - 2x - 6 = \boxed{x^2 + 4x + 3}$$

$$(f \circ f)(x) = f(f(x)) = f(x+3)$$

$$= (x+3) + 3 = \boxed{x+6}$$

$$(g \circ g)(x) = g(g(x)) = g(x^2 - 2x)$$

$$= (x^2 - 2x)^2 - 2(x^2 - 2x)$$

$$= (x^2 - 2x)(x^2 - 2x) - 2x^2 + 4x$$

$$= x^4 - 2x^3 - 2x^3 + 4x^2 - 2x^2 + 4x$$

$$= \boxed{x^4 - 4x^3 + 2x^2 + 4x}$$

Exercise 5.6

Let f and g be the functions defined by the table below. Complete the table by performing the indicated operations.

	x	1	2	3	4	5	6	7
	$f(x)$	4	5	7	0	-2	6	4
	$g(x)$	6	-8	5	2	9	11	2
a)	$f(x) + 3$							
b)	$4g(x) + 5$							
c)	$g(x) - 2f(x)$							
d)	$f(x + 3)$							

Sol

a)

x	1	2	3	4	5	6	7
$f(x)$	4	5	7	0	-2	6	4
$f(x)+3$	7	8	10	3	1	9	7

$$\begin{aligned}
 &\uparrow \quad f(2)+3 \\
 &f(1)+3 = 4+3 \\
 &\quad \quad \quad = 7 \\
 &\quad \quad \quad \parallel \\
 &\quad \quad \quad 4+3=7
 \end{aligned}$$

b)

x	1	2	3	4	5	6	7
$g(x)$	6	-8	5	2	9	11	2
$4g(x)+5$	29	-27	25	13	41	49	13

$$\begin{aligned}
 &\uparrow \quad 4 \cdot g(2)+5 \\
 &4 \cdot g(1)+5 \rightarrow
 \end{aligned}$$

c)

x	1	2	3	4	5	6	7
$f(x)$	4	5	7	0	-2	6	4
$g(x)$	6	-8	5	2	9	11	2

$$g(x) - 2f(x) \quad \begin{array}{|c|c|c|c|c|c|c|c|} \hline -2 & -18 & -9 & 2 & 13 & -1 & -6 \\ \hline \end{array}$$

$$\begin{array}{l}
 g(1) - 2f(1) \quad \uparrow \quad g(2) - 2f(2) \\
 = 6 - 2 \cdot 4 = -2 \quad \quad -8 - 2 \cdot 5 \\
 \quad \quad \quad \quad \quad \quad = -18
 \end{array}$$

d)

x	1	2	3	4	5	6	7
$f(x)$	4	5	7	0	-2	6	4

$$f(x+3) \quad \begin{array}{|c|c|c|c|c|c|c|c|} \hline 0 & -2 & 6 & 4 & \text{undefined} & \text{undef.} & \text{undef.} \\ \hline \end{array}$$

$$\begin{array}{l}
 f(1+3) \\
 = f(4) = 0
 \end{array}$$

$$\begin{array}{l}
 f(5+3) \\
 = f(8) \\
 = \text{undefined}
 \end{array}$$

Exercise 5.7

Let f and g be the functions defined by the table below. Complete the table by composing the given functions.

	x	1	2	3	4	5	6
	$f(x)$	3	1	2	5	6	3
	$g(x)$	5	2	6	1	2	4
a)	$(g \circ f)(x)$						
b)	$(f \circ g)(x)$						
c)	$(f \circ f)(x)$						
d)	$(g \circ g)(x)$						

Sol

a)

x	1	2	3	4	5	6
$f(x)$	3	1	2	5	6	3
$g(x)$	5	2	6	1	2	4

$(g \circ f)(x)$ 6 5 1 2 4 6

$$\begin{aligned}
 (g \circ f)(1) &= g(f(1)) = g(3) = 6 \\
 (g \circ f)(2) &= g(f(2)) = g(1) = 5 \\
 (g \circ f)(3) &= g(f(3)) = g(2) = 1 \\
 (g \circ f)(4) &= g(f(4)) = g(5) = 2 \\
 (g \circ f)(5) &= g(f(5)) = g(6) = 4 \\
 (g \circ f)(6) &= g(f(6)) = g(3) = 6
 \end{aligned}$$

b)

x	1	2	3	4	5	6
$f(x)$	3	1	2	5	6	3
$g(x)$	5	2	6	1	2	4

$(f \circ g)(x)$

$$\begin{aligned}
 (f \circ g)(1) &= f(g(1)) = f(5) = 6 \\
 (f \circ g)(2) &= f(g(2)) = f(2) = 1 \\
 (f \circ g)(3) &= f(g(3)) = f(6) = 3 \\
 (f \circ g)(4) &= f(g(4)) = f(1) = 3 \\
 (f \circ g)(5) &= f(g(5)) = f(2) = 1 \\
 (f \circ g)(6) &= f(g(6)) = f(4) = 5
 \end{aligned}$$

c)

x	1	2	3	4	5	6
$f(x)$	3	1	2	5	6	3

$(f \circ f)(x)$

$$\begin{aligned}
 (f \circ f)(1) &= f(f(1)) = f(3) = 2 \\
 (f \circ f)(2) &= f(f(2)) = f(1) = 3 \\
 (f \circ f)(3) &= f(f(3)) = f(2) = 1 \\
 (f \circ f)(4) &= f(f(4)) = f(5) = 6 \\
 (f \circ f)(5) &= f(f(5)) = f(6) = 3 \\
 (f \circ f)(6) &= f(f(6)) = f(3) = 2
 \end{aligned}$$

(d)

x	1	2	3	4	5	6
$g(x)$	5	2	6	1	2	4

$(g \circ g)(x)$

$(g \circ g)(1) = g(g(1)) = g(5) = 2$

$(g \circ g)(2) = g(g(2)) = g(2) = 2$

$(g \circ g)(3) = g(g(3)) = g(6) = 4$

$(g \circ g)(4) = g(g(4)) = g(1) = 5$

$(g \circ g)(5) = g(g(5)) = g(2) = 2$

$(g \circ g)(6) = g(g(6)) = g(4) = 1$

